*.cha (legacy)

Purpose

This page documents the legacy channel input files: `channel.cha` (original reach pointer), `channel-lte.cha` (SWAT+ deg routing pointer), `hyd-sed-lte.cha` (combined hydrology and sediment table used by the deg routine), `initial.cha` (initial-state pointer), and `temperature.cha` (water temperature parameters). All file names are set in `type input_cha` in `src/input_file_module.f90`. Each file is independently optional: if the slot in `file.cio` is `null` or the file is missing, the reader allocates an empty array and skips on.

The three companion tables that these pointer files reference (`hydrology.cha`, `sediment.cha`, `nutrients.cha`) are documented on the sd_channel page.

Source

- File names: src/input_file_module.f90, `type input_cha`
- Readers:
  - src/ch_read.f90 (`channel.cha`)
  - src/sd_channel_read.f90 (`channel-lte.cha`)
  - src/sd_hydsed_read.f90 (`hyd-sed-lte.cha`)
  - src/ch_read_init.f90 (`initial.cha`)
  - src/ch_read_temp.f90 (`temperature.cha`)
- Type definitions: src/channel_data_module.f90, src/sd_channel_module.f90

Every reader skips two header lines, then reads one record per channel definition.

channel.cha

Pointer file for the original SWAT-style reach routing. Each record names a channel and points to entries in `initial.cha`, `hydrology.cha`, `sediment.cha`, and `nutrients.cha`. Read into `ch_dat_c` (`type channel_data_char_input`).

#	Column	Type	Description
1	id	int	record number (echoed back)
2	name	char(16)	channel name
3	init	char(16)	name of an `initial.cha` entry, or `null`
4	hyd	char(16)	name of a `hydrology.cha` entry, or `null`
5	sed	char(16)	name of a `sediment.cha` entry, or `null`
6	nut	char(16)	name of a `nutrients.cha` entry, or `null`

After reading, the reader resolves each character pointer to an integer index by scanning `ch_init`, `ch_hyd`, `ch_sed`, and `ch_nut`. Missing names are written to the diagnostics file (unit 9001) with messages like `"<name> not found (hydrology.cha)"`. The reader does not abort; the index is left at 0 and downstream code uses the default record.

No example dataset shipping with SWAT+ uses `channel.cha`. The two reference datasets at `refdata/Osu_1hru` and (if present) `refdata/Ames_sub1` configure this slot as `null` in `file.cio` and use `channel-lte.cha` instead.

channel-lte.cha

Pointer file for SWAT+ deg routing. Listed under `chandeg.con`. Each record points to entries in `initial.cha`, `hyd-sed-lte.cha`, an optional sediment table, and `nutrients.cha`. Read into `sd_dat` (`type swatdeg_datafiles`).

#	Column	Type	Description
1	id	int	record number used to position the array element
2	name	char(16)	channel name
3	cha_ini	char(16)	name of an `initial.cha` entry, or `null`
4	cha_hyd	char(16)	name of a `hyd-sed-lte.cha` entry, or `null`
5	cha_sed	char(16)	name of a sediment-nutrient entry (resolved against `sed_nut.cha` when present)
6	cha_nut	char(16)	name of a `nutrients.cha` entry, or `null`

The `cha_hyd` column is matched against names in both `sd_chd` (`hyd-sed-lte.cha`) and `sd_chd1` (`sed_nut.cha`). The `cha_sed` column is read but not resolved to an index by `sd_channel_read`; sediment behaviour follows the `hyd-sed-lte.cha` parameters.

Example from `refdata/Osu_1hru/channel-lte.cha`:

```
channel-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                      cha_ini          cha_hyd          cha_sed          cha_nut
       1  cha01                    initcha1         hydcha01             null          nutcha1
       2  cha03                    initcha1         hydcha03             null          nutcha1
       3  cha04                    initcha1         hydcha04             null          nutcha1
```

hyd-sed-lte.cha

Combined hydrology and bank-erosion parameters for the SWAT+ deg routing routine. One record per channel hydrology profile. Read into `sd_chd` (`type swatdeg_hydsed_data`).

#	Column	Type	Units	Default	Description
1	name	char(25)	none	“”	parameter set name (referenced by `channel-lte.cha`)
2	order	int	none	0	stream order
3	chw	real	m	0.0	channel width
4	chd	real	m	0.0	channel depth
5	chs	real	m/m	0.0	channel slope
6	chl	real	km	0.0	channel length
7	chn	real	none	0.0	channel Manning’s n
8	chk	real	mm/hr	0.0	channel bottom hydraulic conductivity
9	bank_exp	real	none	0.0	bank erosion exponent (column `erod_fact` in editor output)
10	cov	real	0..1	0.0	channel cover factor
11	sinu	real	none	0.0	sinuosity (channel length / straight-line length)
12	vcr_coef	real	none	0.0	critical velocity coefficient (column `eq_slp` in editor output)
13	d50	real	mm	0.0	median sediment diameter
14	ch_clay	real	percent	0.0	clay percent of bank and bed
15	carbon	real	percent	0.0	carbon percent of bank and bed
16	ch_bd	real	t/m^3	0.0	dry bulk density
17	chss	real	none	0.0	channel side slope
18	bankfull_flo	real	m^3/s	0.0	bank-full flow rate (column `bed_load` in editor output)
19	fps	real	m/m	0.000001	flood plain slope
20	fpn	real	none	0.1	flood plain Manning’s n
21	n_conc	real	mg/kg	0.0	nitrogen concentration in channel bank
22	p_conc	real	mg/kg	0.0	phosphorus concentration in channel bank
23	p_bio	real	fraction	0.0	fraction of bank P that is bioavailable

Example from `refdata/Osu_1hru/hyd-sed-lte.cha`:

```
hyd-sed-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                          order            wd            dp           slp           len          mann
hydcha01                          4      27.17726       0.99164       0.00146       3.42760       0.05000
```

Editor column names differ from source field names. The reader uses position only.

initial.cha

Initial state pointer file. Each record names a set of initial-condition files for organic-mineral mass, pesticides, pathogens, heavy metals, and salts. Read into `ch_init` (`type channel_init_datafiles`).

#	Column	Type	Description
1	name	char(16)	initial-state set name
2	org_min	char(16)	initial organic-mineral input file name, or `null`
3	pest	char(16)	initial pesticide input file name, or `null`
4	path	char(16)	initial pathogen input file name, or `null`
5	hmet	char(16)	initial heavy metals input file name, or `null`
6	salt	char(16)	initial salt input file name, or `null`

The reader uses unformatted list-directed reads with no leading integer id column. The record is read directly into the derived type, so the first token on each line is the `name`.

Example from `refdata/Osu_1hru/initial.cha`:

```
initial.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        org_min              pest              path              hmet              salt
initcha1                    no_init              null              null              null              null
```

A separate file `initial.cha_cs` is read by `ch_read_init_cs` for salt and conservative-species initial conditions. The file name is hard-coded in the reader (not in `type input_cha`).

temperature.cha

Water temperature parameters used by the channel temperature routine. Read into `w_temp` (`type water_temperature_data`). The `name` field is `character(len=13)` in the source; longer names will be truncated.

#	Column	Type	Units	Default	Description
1	name	char(13)	none	“”	parameter set name
2	sno_mlt	real	none	1.0	coefficient on snowmelt temperature contribution
3	gw	real	none	1.0	coefficient on groundwater temperature contribution
4	sur_lat	real	none	1.0	coefficient on surface and lateral flow temperature contribution
5	sno_lag	real	days	2.0	air temperature lag to snowmelt (1..3)
6	gw_lag	real	days	250.0	air temperature lag to groundwater flow (200..365)
7	surf_lag	real	days	5.0	air temperature lag to surface runoff (2..5)
8	lat_lag	real	days	10.0	air temperature lag to lateral flow (5..10)
9	lat_lag_coef	real	none	0.75	lateral air-lag coefficient
10	surf_lag_coef	real	none	0.75	surface air-lag coefficient (also used for snow)
11	gw_lag_coef	real	none	0.75	groundwater air-lag coefficient
12	hex_coef1	real	none	0.75	dew point calibration
13	hex_coef2	real	none	1.5	channel geometry calibration
14	sf_on	int	none	0	shade factor file activation. `1` = read from file, `0` = use value from calibration file
15	ssff	real	0..1	0.5	shade factor value (used when `sf_on = 0`)

No example dataset in `refdata/` contains `temperature.cha`. The slot is `null` in `file.cio` for the included reference datasets.

Related files

- `file.cio`. The `channel` line selects which of these files are read.
- `chandeg.con`. Connectivity file that references `channel-lte.cha` entries by name.
- `channel.con`. Connectivity file that references `channel.cha` entries by name.
- sd_channel companion tables (`hydrology.cha`, `sediment.cha`, `nutrients.cha`).